

Probabilistic Dynamic Causal Graphs for Root Cause Analysis with Distributional Forecasting

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Abstract. Root cause analysis (RCA) in industrial multivariate time series aims to identify the variable responsible for an emerging anomaly, but this remains challenging under nonlinear, time-varying, and uncertain sensor dependencies. We propose a probabilistic dynamic RCA framework that infers time-varying lagged causal coefficients and regularizes their temporal evolution using an Ornstein–Uhlenbeck (OU)-based dynamical prior. To improve causality-aware forecasting, the framework separates mean prediction from variance calibration through two-stage learning and suppresses unreliable causal edges with a Signal-to-Noise Ratio (SNR)-based gate. At inference time, anomalous intervals are detected using negative log-likelihood (NLL), and root-cause variables are ranked by combining uncertainty-normalized residual evidence with structural deviation from normal causal-strength patterns. Experiments on synthetic and real-world industrial benchmarks show that the proposed framework improves both causal discovery and RCA performance through probabilistic modeling of time-varying causal structure.

Keywords: Root Cause Analysis · Causality · Ornstein–Uhlenbeck process · Multivariate Time Series.

1 Introduction

In industrial multivariate time series, RCA is a fundamental task that aims to identify the variable that actually caused an anomaly. Industrial systems consist of numerous sensors and control variables that interact in complex ways, and a disturbance originating from a single variable can rapidly propagate to others [10]. Therefore, accurately identifying the root-cause sensor is critically important, as it reduces unnecessary inspections and responses while enabling timely intervention at the true fault location [24]. Causal discovery has thus been regarded as a promising approach for RCA, as it provides an intermediate step

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of predicting directional influences and dependencies among variables [6]. However, existing methods suffer from an important limitation. Many approaches assume a fixed causal graph over the entire time horizon, or summarize potentially time-varying relationships using a single deterministic scalar coefficient [4]. In real industrial processes, however, the strength and direction of inter-sensor interactions can change over time due to operating mode transitions, control interventions, disturbances, and environmental variations [22]. In such settings, compressing evolving causal relationships into static scalar values makes it difficult to capture the true structure at a given moment and can blur the distinction between causes and effects.

This work proposes a probabilistic RCA framework that unifies time-varying causal structure with prediction-based diagnosis. The proposed framework infers a distribution over the causal relationship matrix at each time step and directly incorporates it into a Vector Autoregressive (VAR) [12] style forecasting equation to produce the predictive mean for the next step. In doing so, it parameterizes each variable individually to capture nonlinearity in which the causal coefficient matrix directly contributes to prediction. Furthermore, because causal coefficients in time series tend to evolve smoothly over time, we introduce a dynamical prior based on the Ornstein–Uhlenbeck (OU) process [21] to encourage temporal continuity in the causal structure. Training follows a two-stage strategy that separates the estimation of the predictive mean and variance. In the first stage, the model prioritizes learning the mean predictor and the dynamic causal structure, preventing excessive variance growth from masking mean prediction errors. In this stage, the OU prior and sparsity regularization help stabilize causal relationship estimation. In the second stage, the predictive mean is fixed and only the variance is learned, so that the resulting uncertainty reflects both prediction reliability and temporal and variable-wise variability. During inference, anomalous intervals are detected using a negative log-likelihood-based score normalized by the estimated uncertainty, and root-cause sensors are ranked by aggregating extreme-value patterns in variable-wise prediction residuals within the detected intervals.

2 Related Work

2.1 Causal Discovery

Causal discovery in multivariate time series is commonly formulated under the framework of Granger causality [2], where the past of one variable is considered causal if it improves the prediction of another variable [11]. Classical approaches based on VAR models have been widely studied due to their interpretability, since causal relations can be directly inferred from regression coefficients [1]. However, most VAR-based methods assume linear and time-invariant relationships, which limits their applicability in complex systems that exhibit nonlinear dynamics and evolving dependencies [20].

To address these limitations, recent studies have proposed deep learning-based causal discovery methods. For instance, TCDF [16] employs attention-

based convolutional neural networks to discover causal relationships and associated time delays in multivariate time series. CUTS [5] further extends neural Granger causal discovery by leveraging graph neural networks to infer causal graphs from irregular or partially observed time-series data. More recently, several works have explored integrating causal discovery with root cause analysis. AERCA [6] combines Granger causal discovery with anomaly root cause identification by modeling the distribution of exogenous variables and detecting deviations under abnormal conditions. In addition, UnCLE [3] focuses on discovering time-varying causal dependencies, enabling dynamic causal graphs that capture evolving interactions among variables.

2.2 Root Cause Analysis

RCA aims to identify the underlying variables that initiate anomalous events in complex systems. Early approaches to RCA in multivariate time series primarily relied on per-variable forecasting or reconstruction errors, where variables with the largest deviations are considered potential root causes. While such approaches are simple and sensitive to anomalies, they often fail to distinguish the true initiating variables from downstream responses caused by anomaly propagation [23]. To address this limitation, recent studies have proposed graph-based RCA methods that explicitly model dependencies among variables. For example, RCD [7], CIRCA [9], and RCG [8] utilize dependency or causal graphs to trace anomaly propagation paths and identify upstream candidates as potential root causes. More recently, AERCA [6] integrates Granger causal discovery with RCA by modeling anomalies as interventions on exogenous variables. Specifically, it learns the normal distribution of exogenous variables and identifies root causes as those that significantly deviate from their normal regimes under abnormal conditions.

3 Methods

3.1 Preliminary

Granger Causality Granger causality defines causality in predictive terms: variable j is said to Granger-cause variable i if the past of j improves prediction of the future of i beyond what is already explained by the remaining variables. A classical VAR formulation is as follows:

$$x_i^t = c_i + \sum_{k=1}^K \sum_{j=1}^N A_{ij}^{(k)} x_j^{(t-k)} + \epsilon_i^t, \quad (1)$$

where, N is the number of variables, K is the maximum lag, $A_{ij}^{(k)}$ is the k -lag influence of variable j on variable i , and ϵ_i^t is a noise term. The directed edge $j \rightarrow i$ is absent when $A_{ij}^{(k)} = 0$ for all lags. Our method retains this prediction-based interpretation, but replaces static deterministic coefficients with time-varying approximate posterior distributions.

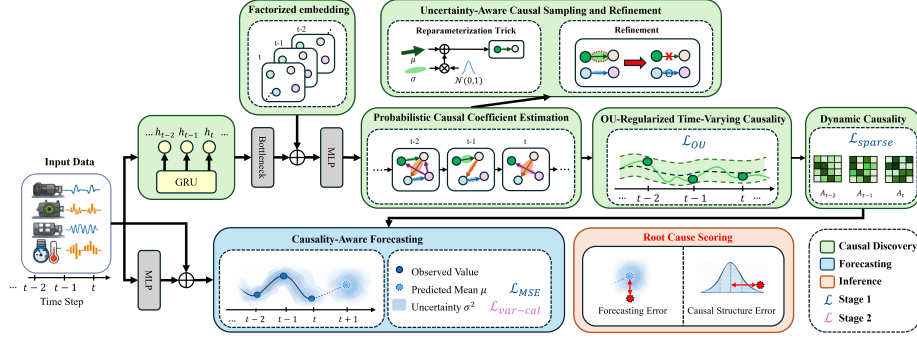


Fig. 1. Overview of the proposed framework.

Ornstein-Uhlenbeck (OU) Process The OU process is a continuous-time stochastic process with mean-reverting dynamics and stochastic fluctuations. These properties make it a widely used model for temporally evolving variables in continuous-time settings [15]. The continuous time OU process is defined as follows:

$$dX_t = \theta(\mu - X_t) dt + \sigma dW_t, \quad (2)$$

where, X_t is the latent state, μ is the long-run mean, $\theta > 0$ is the mean-reversion rate, σ is the diffusion scale, and W_t is standard Brownian motion. For a fixed sampling interval, the induced discrete-time process is an AR(1)-type Gaussian transition. In the zero-mean case used in this paper, this yields a transition of the form

$$X_t \mid X_{t-1} \sim \mathcal{N}(\rho X_{t-1}, Q), \quad (3)$$

with $\rho \in (0, 1)$ and $Q > 0$. This discrete time form is used as the temporal prior over causal coefficients in the proposed model.

3.2 Problem Formulation

Let $x^{1:T} = \{x^1, \dots, x^T\}$ be a multivariate time series with $x^t \in \mathbb{R}^N$, and let $t_0 = K$ be the first valid time index for a length- K history window. For each valid time index $t \in \{t_0, \dots, T-1\}$, we define the lagged history window

$$X_{\text{hist}}^{(t)} = [x^t, x^{t-1}, \dots, x^{t-K+1}] \in \mathbb{R}^{N \times K}, \quad (4)$$

and aim to predict the next observation x^{t+1} . We denote the sequence of history windows available up to time t as the following.

$$\mathcal{H}_t = \left\{ X_{\text{hist}}^{(\tau)} \right\}_{\tau=t_0}^t \quad (5)$$

At the same time, we infer a time-varying lagged causal coefficient $A_t \in \mathbb{R}^{N \times N \times K}$, where $a_{j,i,t}^{(k)}$ denotes the influence from source variable i at lag k to target variable j at time t . Throughout this section, the first node index of $a_{j,i,t}^{(k)}$ is the target and the second is the source.

3.3 Probabilistic Causal Coefficient Estimation

The causal discovery module estimates both lag and edge coefficients as random variables rather than deterministic parameters, so that structural uncertainty is represented explicitly. Conditioned on $\mathcal{H}_t$, we use a diagonal Gaussian variational family over the latent coefficients. To extract temporal context, we encode the sequence of valid history windows with a Gated Recurrent Unit (GRU). For a segment containing times $t_0, \dots, T-1$, we compute

$$[h_{t_0}, \dots, h_{T-1}] = \text{GRU}\left([\text{vec}(X_{\text{hist}}^{(t_0)}), \dots, \text{vec}(X_{\text{hist}}^{(T-1)})]\right), \quad (6)$$

where $\text{vec}(\cdot)$ reshapes each history window into a vector and h_t is the GRU hidden state corresponding to time t . Thus, the GRU processes the temporal sequence of window vectors $\text{vec}(X_{\text{hist}}^{(t)})$ across time, rather than the entries inside a single window independently. We then compress h_t through a bottleneck layer,

$$z_t = \phi_{\text{bot}}(h_t), \quad (7)$$

where ϕ_{bot} is a Multi Layer Perceptron (MLP) and z_t is the compressed dynamic representation at time step t . The purpose of the bottleneck is to prevent the forecasting module from overly relying on the hidden state itself, thereby reducing shortcut behavior and encouraging the model to make predictions through the inferred causal coefficients.

To estimate the distribution of each coefficient $a_{j,i,t}^{(k)}$, we combine temporal context with the structural identity of the target-source-lag tuple. We first construct an edge-level embedding

$$e_{j,i,k} = [e_j^{\text{out}} \parallel e_i^{\text{in}} \parallel e_k^{\text{lag}}], \quad (8)$$

where $\parallel$ denotes concatenation, and e_j^{out} , e_i^{in} , and e_k^{lag} are learnable embeddings for the target role, source role, and lag index, respectively. This design preserves directional asymmetry between incoming and outgoing roles and allows different lags to have distinct structural patterns.

The posterior parameters are generated by a factorized parameterization:

$$u_t = W_h z_t, \quad v_{j,i,k} = W_e e_{j,i,k}, \quad (9)$$

$$g_{j,i,k,t} = \psi(u_t + v_{j,i,k}), \quad (10)$$

$$[\mu_{j,i,t}^{(k)}, \ell_{j,i,t}^{(k)}] = W_o g_{j,i,k,t}, \quad (11)$$

where W_h , W_e , and W_o are learnable linear maps, u_t is the projected temporal context, $v_{j,i,k}$ is the projected edge-lag embedding, $g_{j,i,k,t}$ is the fused hidden representation, $\psi(\cdot)$ is a point-wise nonlinearity, and $\ell_{j,i,t}^{(k)}$ denotes the log-variance parameter, so that $(\sigma_{j,i,t}^{(k)})^2 = \exp(\ell_{j,i,t}^{(k)})$. The additive fusion of u_t and $v_{j,i,k}$ provides a shared temporal context while keeping the number of free parameters

substantially smaller than a fully independent parameterization over (j, i, k, t) . Each causal coefficient is then modeled as a Gaussian random variable:

$$a_{j,i,t}^{(k)} \sim \mathcal{N}\left(\mu_{j,i,t}^{(k)}, (\sigma_{j,i,t}^{(k)})^2\right). \quad (12)$$

Accordingly, the full approximate posterior at time t is as follows:

$$q(A_t | \mathcal{H}_t) = \mathcal{N}(\mu_{A,t}, \text{diag}(\sigma_{A,t}^2)), \quad (13)$$

where $\mu_{A,t}$ and $\sigma_{A,t}$ collect the means and standard deviations of all edge-lag coefficients. The diagonal covariance implies conditional independence across edge-lag coefficients given $\mathcal{H}_t$. Throughout the paper, we refer to $q(A_t | \mathcal{H}_t)$ as an amortized approximate posterior: its parameters are produced by the encoder and optimized jointly with task losses and OU prior matching. In this way, the causal graph at each time step is represented as a probabilistic object with element-wise uncertainty.

3.4 Uncertainty-Aware Causal Sampling and Refinement

During training, we apply the reparameterization trick, introduced in [18], to sample causal coefficients from the posterior in a differentiable manner:

$$\tilde{a}_{j,i,t}^{(k)} = \mu_{j,i,t}^{(k)} + \sigma_{j,i,t}^{(k)} \epsilon_{j,i,t}^{(k)}, \quad \epsilon_{j,i,t}^{(k)} \sim \mathcal{N}(0, 1). \quad (14)$$

Where, $\epsilon_{j,i,t}^{(k)}$ is Gaussian noise sampled from the standard normal distribution, and $\tilde{a}_{j,i,t}^{(k)}$ is a sampled causal coefficient reflecting posterior uncertainty. At the full-structure level, this can be written as follows:

$$\tilde{A}_t = \mu_{A,t} + \sigma_{A,t} \odot \epsilon_t, \quad (15)$$

where $\odot$ denotes element-wise multiplication and $\tilde{A}_t$ is the sampled causal structure at time step t . The sampled coefficients are used as causal weights in the forecasting mean path, thereby enabling the prediction to reflect both uncertainty and coefficient reliability. To further account for the reliability of each inferred causal coefficient, we introduce a SNR [19] based confidence measure. For each edge-lag coefficient, the SNR is defined as follows:

$$\text{SNR}_{j,i,t}^{(k)} = \frac{|\mu_{j,i,t}^{(k)}|}{\sigma_{j,i,t}^{(k)} + \epsilon}, \quad (16)$$

where ϵ is a small constant for numerical stability. The confidence of an edge (j, i) is then summarized by taking the maximum SNR over the lag dimension. This confidence score is transformed into a soft gate as follows:

$$\gamma_{j,i,t} = \text{sigmoid}\left(\frac{\text{SNR}_{j,i,t} - \tau}{\mathbb{T}}\right), \quad (17)$$

where τ is the SNR threshold and T is a temperature parameter controlling the sharpness of the gate. The resulting gate is shared across all lag coefficients of the same edge and applied as follows:

$$\hat{a}_{j,i,t}^{(k)} = \gamma_{j,i,t} \cdot \tilde{a}_{j,i,t}^{(k)} \quad (18)$$

Through this mechanism, the model suppresses the influence of highly uncertain causal coefficients while preserving reliable ones, leading to a more stable use of the inferred causal structure.

Finally, to summarize the overall causal influence of each edge across all lags, we define the lag-aggregated causal strength as follows:

$$\bar{a}_{j,i,t} = \sqrt{\sum_{k=1}^K \left(\hat{a}_{j,i,t}^{(k)} \right)^2}. \quad (19)$$

Where, $\bar{a}_{j,i,t}$ represents the overall causal strength from variable i to variable j at time step t . This aggregated edge-level score is later used for the root-cause ranking.

3.5 OU-Regularized Time-Varying Causality

To introduce temporal dependence between the causal coefficients, we regularize the varying causal coefficients with a discrete-time OU transition. For each edge-lag coefficient,

$$p\left(a_{j,i,t_0}^{(k)}\right) = \mathcal{N}\left(0, \frac{Q}{1-\rho^2}\right) \quad (20)$$

$$p\left(a_{j,i,t}^{(k)} \mid a_{j,i,t-1}^{(k)}\right) = \mathcal{N}\left(\rho a_{j,i,t-1}^{(k)}, Q\right), \quad t > t_0 \quad (21)$$

where t_0 is the first valid time index, $\rho \in (0, 1)$ controls temporal persistence, and $Q > 0$ is the process-noise variance. Under a factorized prior across edge-lag coefficients,

$$p(A_t \mid A_{t-1}) = \prod_{j=1}^N \prod_{i=1}^N \prod_{k=1}^K p\left(a_{j,i,t}^{(k)} \mid a_{j,i,t-1}^{(k)}\right). \quad (22)$$

Where, Q is a scalar process-noise variance shared across coefficients. Equivalently, if A_t is vectorized into $\text{vec}(A_t) \in \mathbb{R}^{N^2 K}$, the transition prior is a diagonal Gaussian with mean $\rho \text{vec}(A_{t-1})$ and covariance QI , where I is the identity matrix in the vectorized coefficient space. This is the stationary AR(1) form induced by the OU process at discrete time points.

In implementation, we use the previous approximate-posterior mean as a plug-in estimate of the latent state and stop its gradient when constructing the transition center. Hence, for $t > t_0$, the prior used in the Kullback–Leibler (KL) term and the resulting prior-matching penalty is defined as

$$p(A_t \mid \text{sg}(\mu_{A,t-1})) = \mathcal{N}(\rho \text{sg}(\mu_{A,t-1}), QI) \quad (23)$$

$$\mathcal{L}_{\text{OU}} = D_{\text{KL}}(q(A_{t_0} | \mathcal{H}_{t_0}) \parallel p(A_{t_0})) + \sum_{t=t_0+1}^{T-1} D_{\text{KL}}(q(A_t | \mathcal{H}_t) \parallel p(A_t | \text{sg}(\mu_{A,t-1}))). \quad (24)$$

Where, $D_{\text{KL}}(\cdot \parallel \cdot)$ denotes KL divergence, $\text{sg}(\cdot)$ denotes stop-gradient, and $\mu_{A,t-1}$ is the approximate-posterior mean at time $t-1$. As both $q(\cdot)$ and $p(\cdot)$ are diagonal Gaussians, each KL term is available in closed form. This regularizer discourages abrupt noise-driven changes while still allowing meaningful structural evolution when supported by the data.

3.6 Causality-Aware Forecasting

To use the inferred causal coefficients directly in prediction, we design a nonlinear VAR-style forecasting module. Let $\phi_{\text{map}}(x) = x + \text{MLP}(x)$ be a shared scalar-to-scalar nonlinear mapping applied independently to each lagged observation. The predictive mean for target variable j is

$$\mu_{j,t+1} = \sum_{i=1}^N \sum_{k=1}^K \hat{a}_{j,i,t}^{(k)} \phi_{\text{map}}(x_i^{t-k+1}) + c_j \quad (25)$$

where $\hat{a}_{j,i,t}^{(k)}$ is the effective gated coefficient defined above and c_j is a bias term. This formulation captures local nonlinearity through $b(\cdot)$ while forcing all cross-variable influence to pass through the causal coefficients.

Predictive uncertainty is modeled separately from the mean path. Let $x_{j,t:t-K+1} = [x_j^t, x_j^{t-1}, \dots, x_j^{t-K+1}] \in \mathbb{R}^K$ denote the recent history of variable j . The variance head for variable j uses only this univariate history:

$$\sigma_{j,t+1}^2 = \text{softplus}(f_j(x_{j,t:t-K+1})) + \varepsilon \quad (26)$$

where $f_j : \mathbb{R}^K \rightarrow \mathbb{R}$ is the variance head for variable j . This decoupling prevents cross-variable shortcuts in the variance pathway and keeps the causal coefficients as the unique route by which inter-variable effects influence the predictive mean.

3.7 Training Objective

As aforementioned, the model can reduce loss lazily by inflating variance. This weakens the role of the causal path in forecasting, therefore requiring a two-stage training scheme.

Stage 1: Causal Discovery and Mean Forecasting In Stage 1, the model learns the dynamic causal structure, the parameters of the approximate posterior, and the predictive mean. The mean forecasting loss is

$$\mathcal{L}_{\text{mse}} = \frac{1}{mN} \sum_{t=1}^m \|x^{t+1} - \mu_{t+1}\|_2^2, \quad (27)$$

where m is the number of valid prediction steps in the current segment.

To discourage dense graphs, we impose an edge-wise sparsity penalty on the lag-aggregated strength:

$$\mathcal{L}_{\text{sparse}} = \frac{1}{N^2} \sum_{j=1}^N \sum_{i=1}^N \sqrt{\frac{1}{m} \sum_{t=1}^m \bar{a}_{j,i,t}^2 + \varepsilon}. \quad (28)$$

Combining these terms with the OU regularizer, Stage 1 minimizes

$$\mathcal{L}_{\text{S1}} = \mathcal{L}_{\text{mse}} + \lambda_{\text{sp}} \mathcal{L}_{\text{sparse}} + \lambda_{\text{ou}} \mathcal{L}_{\text{OU}}. \quad (29)$$

Where, $\lambda_{\text{sp}} > 0$ and $\lambda_{\text{ou}} > 0$ are the sparsity and OU-regularization weights, respectively. This objective should be interpreted as task-driven amortized inference with prior matching: the approximate posterior is learned through forecasting utility and OU regularization.

Stage 2: Variance Calibration In Stage 2, we freeze the mean path learned in Stage 1 and optimize only the predictive variance. Let the residual for variable j at time $t+1$ be $r_{j,t+1} = x_{j,t+1} - \mu_{j,t+1}$. The main variance-calibration term is

$$\mathcal{L}_{\text{var-cal}} = \frac{1}{mN} \sum_{t=1}^m \sum_{j=1}^N (\log \sigma_{j,t+1}^2 - \log (r_{j,t+1}^2 + \varepsilon))^2. \quad (30)$$

The Stage 2 objective is therefore $\mathcal{L}_{\text{S2}} = \mathcal{L}_{\text{var-cal}}$. No additional variance regularizers are used in this stage.

3.8 Root Cause Scoring

During inference, we first obtain the predictive mean and variance from the forecasting module. For each anomalous time step t , root-cause ranking is performed by combining a forecasting error-based score and a structural-deviation score. First, the forecasting error of variable i at time step t is defined as follows:

$$r_{i,t} = \frac{|x_{i,t} - \mu_{i,t}|}{\sqrt{\sigma_{i,t}^2 + \epsilon}}, \quad (31)$$

where $\mu_{i,t}$ and $\sigma_{i,t}^2$ denote the predictive mean and variance, respectively, and ϵ is a small positive constant for numerical stability. Since this definition explicitly incorporates forecasting uncertainty, errors in highly uncertain regions are reflected more conservatively, whereas errors under confident predictions receive larger scores. Next, the structural deviation is defined as the test time causal strength deviation from the time-varying causal strength, represented as a normal distribution learned from the training data. We first compute the lag-aggregated causal strength of edge ($i \rightarrow j$) at time step t as follows:

$$\bar{a}_{j,i,t} = \sqrt{\sum_{k=1}^K \left(\hat{a}_{j,i,t}^{(k)} \right)^2}, \quad (32)$$

where $\hat{a}_{j,i,t}^{(k)}$ denotes the causal coefficient used for scoring. Let $\mu_{j,i}^{\text{ref}}$ and $\sigma_{j,i}^{\text{ref}}$ denote the mean and standard deviation of the time-varying causal distribution. The structural deviation of the corresponding edge is then defined as follows:

$$d_{j,i,t} = \frac{|\bar{a}_{j,i,t} - \mu_{j,i}^{\text{ref}}|}{\sigma_{j,i}^{\text{ref}} + \epsilon}. \quad (33)$$

Thus, $d_{j,i,t}$ measures how far the current causal strength deviates from the normal causal distribution. The variable-wise structural score of source variable i is obtained by averaging the deviations of its outgoing edges:

$$d_{i,t} = \frac{1}{N} \sum_{j=1}^N d_{j,i,t}. \quad (34)$$

Finally, the forecast-based score and the structural score are normalized separately and linearly combined to compute the final root-cause score:

$$\tilde{r}_{i,t} = \frac{r_{i,t}}{\sum_{n=1}^N r_{n,t} + \epsilon}, \quad \tilde{d}_{i,t} = \frac{d_{i,t}}{\sum_{n=1}^N d_{n,t} + \epsilon}. \quad (35)$$

Where, n is an index over all variables, and ϵ is a small positive constant. The final root-cause score at time step t is defined as follows:

$$\text{RCA}_{i,t} = \tilde{r}_{i,t} + \tilde{d}_{i,t}, \quad (36)$$

and the root-cause sensor is determined by ranking $\text{RCA}_{i,t}$ in descending order at each detected anomalous time step.

Table 1. Dynamic causal discovery performance on ND8 (mean $\pm$ std).

Model	ND8	
	AUC-ROC	AUC-PRC
GVAR[13]	0.720 $\pm$ 0.016	0.225 $\pm$ 0.020
UnCLe[3]	0.918 $\pm$ 0.005	0.613 $\pm$ 0.055
Ours	0.930 $\pm$ 0.008	0.647 $\pm$ 0.016

4 Experiments

4.1 Experimental Setup

We evaluate the proposed framework on both synthetic and real-world multivariate time-series benchmarks. For causal discovery, we use the synthetic *Linear* and *Nonlinear* datasets generated following [6], both of which provide ground-truth causal structures under controlled linear and nonlinear temporal dependencies. To further evaluate dynamic causal discovery, we adopt the *ND8* benchmark constructed according to [3]. *ND8* is a simulated nonlinear dynamic dataset with eight variables, where the causal directions of selected variable pairs periodically switch over time. This benchmark is therefore used to assess whether the proposed framework can capture time-varying causal structures beyond a single static graph. For root cause analysis,

we use *SWaT* and *MSDS*, two real-world industrial datasets with labeled anomalous events and root-cause annotations. In all datasets, the model is trained on normal sequences, while anomalous windows are reserved for evaluation.

We evaluate causal discovery using *AUROC* and *AUPRC*, which measure how well the inferred edge strengths recover ground-truth causal edges. For RCA, we report *Accuracy at K* (AC@K), following the protocol for root cause identification. Let $\mathcal{V}^{RC}$ denote the set of ground-truth root-cause variables and $\mathcal{R}_K$ denote the set of top- K variables ranked by the RCA score. AC@K is defined as

$$\text{AC@K} = \frac{|\mathcal{V}^{RC} \cap \mathcal{R}_K|}{\min(K, |\mathcal{V}^{RC}|)}. \quad (37)$$

Thus, AC@K measures whether the ground-truth root-cause variables are included among the top- K ranked candidates. We report AC@1, AC@3, AC@5, and AC@10, where K denotes the number of top-ranked candidate variables considered.

Table 2. Causal discovery performance across simulated datasets (mean $\pm$ std).

Model	Linear		Nonlinear	
	AUC-ROC	AUC-PRC	AUC-ROC	AUC-PRC
VAR[12]	0.999 $\pm$ 0.001	0.999 $\pm$ 0.001	0.662 $\pm$ 0.137	0.518 $\pm$ 0.181
TCDF[16]	0.934 $\pm$ 0.056	0.917 $\pm$ 0.81	0.648 $\pm$ 0.060	0.332 $\pm$ 0.071
GVAR[13]	0.986 $\pm$ 0.028	0.981 $\pm$ 0.040	0.683 $\pm$ 0.091	0.560 $\pm$ 0.145
Cuts[5]	0.865 $\pm$ 0.095	0.812 $\pm$ 0.066	0.561 $\pm$ 0.025	0.268 $\pm$ 0.010
AERCA[6]	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.999 $\pm$ 0.001	0.998 $\pm$ 0.006
Ours	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000

4.2 Experiment Results

We first evaluate whether the proposed framework can recover dynamic causal relationships. Table 1 shows that our method achieves the best AUROC and AUPRC among the compared methods in ND8, and Table 2 shows that it also performs strongly on the Linear and Nonlinear synthetic benchmarks. These results indicate that the proposed formulation remains competitive across both controlled non-dynamic simulations and the dynamic benchmark.

Table 3 summarizes the RCA performance in terms of AC@K. The proposed method achieves the strongest overall performance across the evaluated datasets, surpassing the baselines by a substantial margin. In particular, AERCA performs well on the synthetic dataset but suffers a marked degradation on real-world benchmarks, while RCG shows competitive performance on SWaT only and fails to maintain this trend on the remaining datasets. By contrast, our method yields

Table 3. Root cause analysis performance (AC@K) across datasets (mean $\pm$ std). The first two datasets are simulated and the remaining two are real industrial datasets.

Dataset	Method	AC@1	AC@3	AC@5	AC@10
Linear	RCD[7]	0.500 $\pm$ 0.450	0.817 $\pm$ 0.149	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	CIRCA[9]	0.600 $\pm$ 0.457	0.800 $\pm$ 0.296	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	RCG[8]	0.646 $\pm$ 0.140	0.794 $\pm$ 0.075	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	AERCA[6]	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	Ours	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Nonlinear	RCD[7]	0.600 $\pm$ 0.440	0.750 $\pm$ 0.317	0.880 $\pm$ 0.156	1.000 $\pm$ 0.000
	CIRCA[9]	0.700 $\pm$ 0.427	0.717 $\pm$ 0.387	0.835 $\pm$ 0.117	1.000 $\pm$ 0.000
	RCG[8]	0.591 $\pm$ 0.117	0.715 $\pm$ 0.020	0.897 $\pm$ 0.008	1.000 $\pm$ 0.000
	AERCA[6]	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	Ours	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
SWaT[14]	RCD[7]	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.300 $\pm$ 0.104
	CIRCA[9]	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.300 $\pm$ 0.041
	RCG[8]	0.327 $\pm$ 0.104	0.370 $\pm$ 0.088	0.401 $\pm$ 0.043	0.562 $\pm$ 0.041
	AERCA[6]	0.220 $\pm$ 0.104	0.290 $\pm$ 0.088	0.330 $\pm$ 0.043	0.455 $\pm$ 0.041
	Ours	0.241 $\pm$ 0.053	0.517 $\pm$ 0.037	0.621 $\pm$ 0.074	0.758 $\pm$ 0.038
MSDS[17]	RCD[7]	0.412 $\pm$ 0.051	0.573 $\pm$ 0.020	0.984 $\pm$ 0.001	1.000 $\pm$ 0.000
	CIRCA[9]	0.454 $\pm$ 0.238	0.860 $\pm$ 0.138	0.917 $\pm$ 0.080	1.000 $\pm$ 0.000
	RCG[8]	0.527 $\pm$ 0.302	0.591 $\pm$ 0.255	0.647 $\pm$ 0.057	1.000 $\pm$ 0.000
	AERCA[6]	0.395 $\pm$ 0.402	0.908 $\pm$ 0.064	0.974 $\pm$ 0.022	1.000 $\pm$ 0.000
	Ours	0.827 $\pm$ 0.102	0.982 $\pm$ 0.0084	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000

consistently strong results across all datasets, demonstrating superior robustness and generalization.

Figure 2 further supports the structural-deviation term used in the RCA score. In both examples, the anomalous causal strengths lie far from the normal training distribution. This means that anomalies are accompanied not only by forecasting errors but also by clear departures from the normal causal mechanism, which empirically justifies including structural deviation as a complementary RCA signal.

4.3 Uncertainty Experiment Results

Figure 3 examines whether the learned causal-discovery uncertainty reflects edge reliability. In the left panel, the performance of the model, in AUROC, is compared directly with the percentage of uncertain edges included. The causal discovery performance decreases from 0.93 to approximately 0.87 as more low-confidence edges are added. This indicates that uncertain edges hinder model performance. In the right panel, all inferred edges are sorted by predicted uncertainty and divided into four equal-sized quartiles. For each quartile, the mean edge-level classification error is computed as $(1 - p_{\text{active}})$ for a ground-truth active edge and p_{active} for a ground-truth inactive edge, where p_{active} denotes the predicted edge-activation probability. The error increases monotonically from

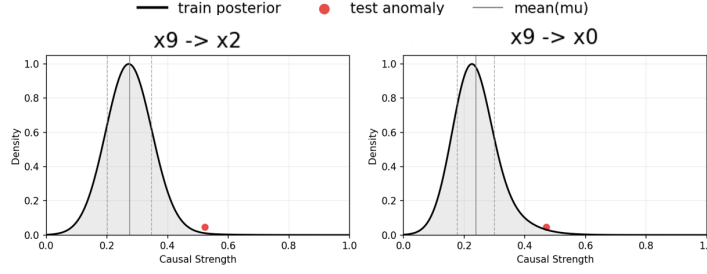


Fig. 2. Structural deviation under anomaly relative to the normal causal distribution. The black curve shows the causal-strength posterior estimated from normal training data, the red point shows the causal strength inferred at anomalous test time, and the gray vertical line denotes the posterior mean μ . The labels $x9 \rightarrow x2$ and $x9 \rightarrow x0$ indicate causal edges from source variable $x9$ to target variables $x2$ and $x0$.

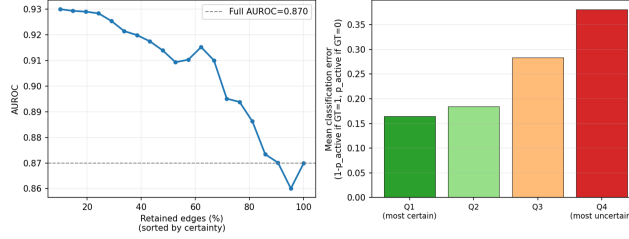


Fig. 3. Uncertainty diagnostics for causal edge reliability on ND8. Left: AUROC evaluated as edges are retained according to predicted certainty. The dashed line denotes the AUROC computed using all inferred edges. Right: mean edge-level classification error computed for four groups of edges formed by uncertainty quartiles.

Q1 to Q4, which further shows that uncertainty is directly correlated with less accurate edges. Together, these diagnostics provide empirical justification for suppressing highly uncertain causal edges through the uncertainty-aware SNR gate introduced in the model.

Figure 4 evaluates whether the forecasting uncertainty estimated by the variance head reflects the difficulty of prediction. In the left panel, samples are sorted from the most certain to the least certain, and the mean absolute forecasting error is shown in relation to the inclusion of an increasing percent of uncertain samples. The error increases steadily as low-confidence samples are added, indicating that the predicted uncertainty is aligned with actual forecasting difficulty. In the right panel, samples are sorted by predictive uncertainty and split into four equal-sized quartiles. The mean absolute error again increases monotonically from Q1 to Q4. These two diagnostics support the interpretation that the forecasting uncertainty is meaningful as a relative confidence measure and provide empirical justification for using uncertainty-normalized forecasting evidence in anomaly scoring and RCA.

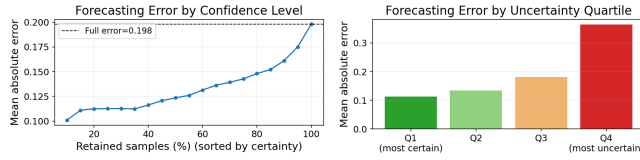


Fig. 4. Forecasting uncertainty diagnostics on SWaT. Left: mean absolute error under certainty-based sample retention. Right: mean absolute error across uncertainty quartiles.

Table 4. Ablation Study on SWaT

Module	AC@1	AC@3	AC@5	AC@10
OU w/o	0.231	0.379	0.448	0.625
NLL w/o	0.206	0.482	0.586	0.655
SNR gate w/o	0.172	0.310	0.379	0.688
$\mathcal{L}_{sparse}$ w/o	0.172	0.448	0.512	0.689
Joint stage1 & 2	0.209	0.379	0.443	0.627
Full	0.241	0.517	0.621	0.758

4.4 Ablation Study

Table 4 shows the ablation results on the SWaT dataset for evaluating the contribution of each design component in the proposed framework. The full model achieves the best performance across all metrics. In contrast, removing any of the major components consistently degrades RCA performance, suggesting that the overall gain comes from the interaction of complementary components rather than any single module. Among the ablated variants, removing the uncertainty-aware SNR gate causes the largest performance drop in the top-ranked metrics. Specifically, AC@3 and AC@5 decrease to 0.310 and 0.379, respectively. This is consistent with the uncertainty analysis in Fig. 3. It suggests that suppressing uncertain edges is critical for preventing noisy or weak causal connections from adversely affecting the RCA ranking.

The two-stage training strategy also plays an essential role. When Stages 1 and 2 are joined together to learn the predictive mean and variance simultaneously, the performance again deteriorates. This supports separating mean learning from variance calibration. The result proves that, when both objectives are optimized at the same time, the model will rely on inflating predictive variance before learning accurate predictive means to lazily reduce its loss value. Removing the OU prior also degrades performance, suggesting that smooth temporal evolution of the causal structure improves RCA reliability. Similarly, the sparsity term is beneficial, as explicitly controlling graph density helps prevent the causal graph from being diffused across many weak edges and enables clearer separation of the dominant causal pathways. Finally, the NLL w/o variant corresponds to a setting that uses only MSE instead of NLL. Its inferior performance shows

that incorporating predictive variance, rather than relying solely on the predictive mean, makes a meaningful contribution to RCA performance by enabling uncertainty-aware scoring.

5 Conclusion

We proposed a probabilistic dynamic causal RCA framework for industrial multivariate time series. The model estimates time-varying causal coefficients as approximate posterior distributions, regularizes their evolution with an OU-inspired prior, and integrates the inferred causal structure into a nonlinear forecasting model. This unified design supports causal discovery, uncertainty-aware forecasting, and improved root-cause localization.

Experiments show strong performance on synthetic and real datasets, with meaningful causal uncertainty, well-calibrated forecasting uncertainty, and improved RCA results. However, the current framework is limited by the smoothness assumption of the OU-based prior, which makes it less suitable for settings with abrupt switch-like causal changes. Handling such discrete structural transitions more effectively remains an important direction for future work.

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Use of Generative AI. A generative AI tool was used only to assist with language polishing, including grammar correction, translation, and improving the readability of author-written text.

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